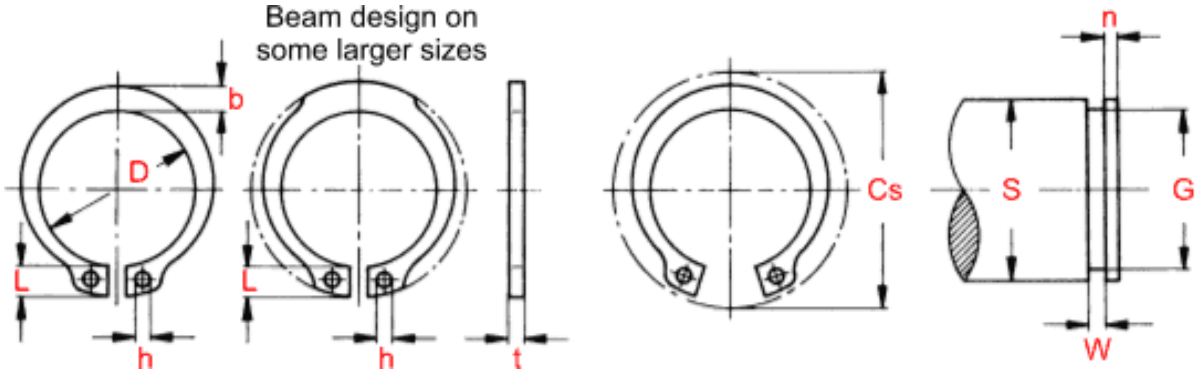
Standard External Circlips



Standard External Circlips: To DIN 471 Metric

The most common form of axially fitted circlips for shafts with grooves. Constructed as curved beams of uniform strength, the radial width is reduced towards the free ends so that a constantly round shape is maintained throughout continuous use. This form of circlip often used in assemblies subjected to strong centrifugal forces, is also secure against high rotational speeds. Lugs with holes are provided for rapid fitting or removal with circlip pliers.

Standard material: Carbon Spring Steel - Standard Finish: Phosphate and Oil.
(Standard circlips are also available in Stainless Steel)

Key to Table Headings and Abbreviations

Al -Axial Load
b -Beam
B -Bore Diameter
Brg-Nominal Dimensions of Bearings
C -Circlip in Groove
Cb -Clearance in Bore
Cm-Circlip on Shaft
Cs -Clearance on Shaft
D -Free Diameter
Fn -Load Carrying Capacity of Groove
Fr -Load Carrying Capacity of Ring
g -Gap Size
G -Groove Diameter
h -Hole Diameter
ins. -Inches
kN -Kilonewtons
lb.f -Pounds force
L -Lug
mm -Millimetres
n -Edge Margin
N -Newtons
r -Radius
Rd -Radial Depth
RPM-Revs per Minute
S -Shaft Diameter
t -Thickness
Tol -Tolerance
W -Groove Width

Most sizes over 170mm are without lugs

Measurements in mm & kN

Shaft	Circlip Dimensions									Groove Dimensions					Code No.
	t	Tol	D	Tol	Cs	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
3	0.40	+0.00 -0.05	2.7	+0.04 -0.15	7.0	0.8	1.9	1.0	0.47	2.8	+0.00 -0.04	0.5	0.3	0.15	EXT0030
4	0.40		3.7		8.6	0.9	2.2	1.0	0.50	3.8	+0.00 -0.048	0.5	0.3	0.20	EXT0040
5	0.60		4.7		10.3	1.1	2.5	1.0	1.00	4.8		0.7	0.3	0.26	EXT0050
6	0.70		5.6		11.7	1.3	2.7	1.2	1.45	5.7		0.8	0.5	0.46	EXT0060
7	0.80		6.5	+0.06 -0.18	13.5	1.4	3.1	1.2	2.60	6.7	+0.00 -0.06	0.9	0.5	0.54	EXT0070
8	0.80		7.4		14.7	1.5	3.2	1.2	3.00	7.6		1.1	0.6	0.81	EXT0080
9	1.00	+0.00 -0.06	8.4		16.0	1.7	3.3	1.2	3.50	8.6		1.1	0.6	0.92	EXT0090
10	1.00		9.3	+0.10 -0.36	17.0	1.8	3.3	1.5	4.00	9.6		1.1	0.6	1.01	EXT0100
11	1.00		10.2		18.0	1.8	3.3	1.5	4.50	10.5	+0.00 -0.11	1.1	0.8	1.40	EXT0110
12	1.00		11.0		19.0	1.8	3.3	1.7	5.00	11.5		1.1	0.8	1.53	EXT0120
13	1.00		11.9		20.2	2.0	3.4	1.7	5.80	12.4		1.1	0.9	2.00	EXT0130
14	1.00		12.9		21.4	2.1	3.5	1.7	6.35	13.4		1.1	0.9	2.15	EXT0140
15	1.00		13.8		22.6	2.2	3.6	1.7	6.9	14.3		1.1	1.1	2.66	EXT0150
16	1.00		14.7		23.8	2.2	3.7	1.7	7.4	15.2		1.1	1.2	3.26	EXT0160
17	1.00		15.7		25.0	2.3	3.8	1.7	8.0	16.2		1.1	1.2	3.46	EXT0170
18	1.20		16.5		26.2	2.4	3.9	2.0	17.0	17.0		1.3	1.5	4.58	EXT0180
19	1.20		17.5		27.2	2.5	3.9	2.0	17.0	18.0		1.3	1.5	4.84	EXT0190
20	1.20		18.5	+0.13 -0.42	28.4	2.6	4.0	2.0	17.1	19.0	+0.00 -0.21	1.3	1.5	5.06	EXT0200
21	1.20		19.5		29.6	2.7	4.1	2.0	16.8	20.0		1.3	1.5	5.36	EXT0210
22	1.20		20.5		30.8	2.8	4.2	2.0	16.9	21.0		1.3	1.5	5.65	EXT0220
23	1.20		21.5		32.0	2.9	4.3	2.0	16.6	22.0		1.3	1.5	5.90	EXT0230
24	1.20		22.2	+0.21 -0.42	33.2	3.0	4.4	2.0	16.1	22.9		1.3	1.7	6.75	EXT0240
25	1.20		23.2		34.2	3.0	4.4	2.0	16.2	23.9		1.3	1.7	7.05	EXT0250
26	1.20		24.2		35.5	3.1	4.5	2.0	16.1	24.9		1.3	1.7	7.34	EXT0260
27	1.20		24.9		36.7	3.1	4.6	2.0	16.4	25.6		1.3	2.1	9.63	EXT0270
28	1.50		25.9		37.9	3.2	4.7	2.0	32.1	26.6		1.6	2.1	10.00	EXT0280
29	1.50		26.9		39.1	3.4	4.8	2.0	31.8	27.6		1.6	2.1	10.37	EXT0290
30	1.50		27.9		40.5	3.5	5.0	2.0	32.1	28.6		1.6	2.1	10.73	EXT0300
31	1.50		28.6		41.5	3.5	5.0	2.5	31.5	29.3		1.6	2.6	13.85	EXT0310
32	1.50		29.6		43.0	3.6	5.2	2.5	31.2	30.3		1.6	2.6	13.85	EXT0320
33	1.50		30.5		44.0	3.7	5.2	2.5	31.6	31.3		1.6	2.6	14.30	EXT0330

34	1.50		31.5	+0.25 -0.50	45.4	3.8	5.4	2.5	31.3	32.3	+0.00 -0.25	1.6	2.6	14.72	EXT0340
35	1.50		32.2		46.8	3.9	5.6	2.5	30.8	33.0		1.6	3.0	17.80	EXT0350
36	1.75		33.2		47.8	4.0	5.6	2.5	49.4	34.0		1.85	3.0	18.33	EXT0360
37	1.75		34.2		49.0	4.1	5.7	2.5	50.0	35.0		1.85	3.0	18.80	EXT0370
38	1.75		35.2		50.2	4.2	5.8	2.5	49.5	36.0		1.85	3.0	19.30	EXT0380
39	1.75		36.0		51.4	4.3	5.9	2.5	49.8	37.0		1.85	3.8	19.90	EXT0390
40	1.75		36.5		52.6	4.4	6.0	2.5	51.0	37.5		1.85	3.8	25.30	EXT0400
41	1.75		37.5		54.1	4.5	6.2	2.5	50.1	38.5		1.85	3.8	26.00	EXT0410
42	1.75		38.5	+0.39 -0.90	55.7	4.5	6.5	2.5	50.0	39.5		1.85	3.8	26.70	EXT0420
43	1.75		39.5		56.7	4.6	6.5	2.5	49.0	40.5		1.85	3.8	27.30	EXT0430
44	1.75		40.5		57.9	4.6	6.6	2.5	48.5	41.5		1.85	3.8	28.00	EXT0440
45	1.75		41.5		59.1	4.7	6.7	2.5	49.0	42.5		1.85	3.8	28.06	EXT0450
46	1.75		42.5		60.1	4.8	6.7	2.5	48.9	43.5		1.85	3.8	29.04	EXT0460
47	1.75		43.5		61.3	4.9	6.8	2.5	49.5	44.5		1.85	3.8	30.00	EXT0470
48	1.75		44.5		62.5	5.0	6.9	2.5	49.4	45.5		1.85	3.8	30.07	EXT0480
50	2.00	+0.00 -0.07	45.8		64.5	5.1	6.9	2.5	73.3	47.0		2.15	4.5	38.00	EXT0500
51	2.00		46.8		65.7	5.2	7.0	2.5	73.2	48.0		2.15	4.5	38.80	EXT0510
52	2.00		47.8		66.7	5.2	7.0	2.5	73.1	49.0		2.15	4.5	39.70	EXT0520
53	2.00		48.8		68.0	5.3	7.1	2.5	72.2	50.0		2.15	4.5	40.40	EXT0530
54	2.00		49.8	+0.46 -1.10	69.0	5.3	7.1	2.5	71.2	51.0	+0.00 -0.30	2.15	4.5	41.20	EXT0540
55	2.00		50.8		70.2	5.4	7.2	2.5	71.4	52.0		2.15	4.5	42.00	EXT0550
56	2.00		51.8		71.6	5.5	7.3	2.5	70.8	53.0		2.15	4.5	42.80	EXT0560
57	2.00		52.8		72.4	5.5	7.3	2.5	70.9	54.0		2.15	4.5	43.70	EXT0570
58	2.00		53.8		73.6	5.6	7.3	2.5	71.1	55.0		2.15	4.5	44.30	EXT0580

Most sizes over 170mm are without lugs

Measurements in mm & kN

Shaft	Circlip Dimensions									Groove Dimensions					Code No.
	t	Tol	D	Tol	Cs	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
60	2.00	+0.00 -0.07	55.8	+0.46 -1.10	75.6	5.8	7.4	2.5	69.2	57.0	+0.00 -0.30	2.15	4.5	46.00	EXT0600
62	2.00		57.8		77.8	6.0	7.5	2.5	69.3	59.0		2.15	4.5	47.50	EXT0620
63	2.00		58.8		79.0	6.2	7.6	2.5	70.2	60.0		2.15	4.5	48.30	EXT0630
65	2.50		60.8		81.4	6.3	7.8	3.0	135.6	62.0		2.65	4.5	49.80	EXT0650
67	2.50		62.5		83.6	6.4	7.9	3.0	136.1	64.0		2.65	4.5	51.30	EXT0670
68	2.50		63.5		84.4	6.5	8.0	3.0	135.9	65.0		2.65	4.5	52.20	EXT0680
70	2.50		65.5		87.0	6.6	8.1	3.0	134.2	67.0		2.65	4.5	53.80	EXT0700
72	2.50		67.5		89.2	6.8	8.2	3.0	131.8	69.0		2.65	4.5	55.30	EXT0720
75	2.50		70.5		92.7	7.0	8.4	3.0	130.0	72.0		2.65	4.5	57.60	EXT0750
77	2.50		72.5		94.9	7.2	8.5	3.0	131.3	74.0		2.65	4.5	59.30	EXT0770
78	2.50		73.5		96.1	7.3	8.6	3.0	131.3	75.0		2.65	4.5	60.00	EXT0780

80	2.50		74.5		98.1	7.4	8.6	3.0	128.4	76.5		2.65	5.3	71.60	EXT0800
82	2.50		76.5		100.3	7.6	8.7	3.0	128.0	78.5		2.65	5.3	73.50	EXT0820
85	3.00		79.5		103.3	7.8	8.7	3.5	215.4	81.5		3.15	5.3	76.20	EXT0850
87	3.00		81.5		105.5	7.9	8.8	3.5	222.2	83.5		3.15	5.3	78.20	EXT0870
88	3.00		82.5		106.5	8.0	8.8	3.5	221.8	84.5		3.15	5.3	79.00	EXT0880
90	3.00		84.5		108.5	8.2	8.8	3.5	217.2	86.5		3.15	5.3	80.8	EXT0900
92	3.00	+0.00 -0.08	86.5		111.0	8.4	9.0	3.5	217.0	88.5	+0.00 -0.35	3.15	5.3	82.7	EXT0920
95	3.00		89.5		114.8	8.6	9.4	3.5	212.2	91.5		3.15	5.3	85.5	EXT0950
97	3.00		91.5		116.8	8.8	9.4	3.5	211.1	93.5		3.15	5.3	87.3	EXT0970
98	3.00		92.5		118.0	9.0	9.5	3.5	208.1	94.5		3.15	5.3	88.2	EXT0980
100	3.00		94.5	+0.54 -1.30	120.2	9.0	9.6	3.5	206.4	96.5		3.15	5.3	90.0	EXT1000
102	4.00		95.0		122.4	9.2	9.7	3.5	482.3	98.0		4.15	6.0	104.60	EXT1020
105	4.00		98.0		125.8	9.3	9.9	3.5	471.8	101.0		4.15	6.0	107.60	EXT1050
108	4.00		101.0		129.0	9.5	10.0	3.5	459.8	104.0		4.15	6.0	111.00	EXT1080
110	4.00		103.0		131.2	9.6	10.1	3.5	457.0	106.0	+0.00 -0.54	4.15	6.0	113.00	EXT1100
112	4.00		105.0		133.7	9.7	10.3	3.5	451.5	108.0		4.15	6.0	115.00	EXT1120
115	4.00		108.0		137.7	9.8	10.6	3.5	438.6	111.0		4.15	6.0	118.20	EXT1150
120	4.00		113.0		143.1	10.2	11.0	3.5	424.6	116.0		4.15	6.0	123.5	EXT1200
125	4.00		118.0		149.0	10.4	11.4	4.0	411.5	121.0		4.15	6.0	128.7	EXT1250
130	4.00		123.0		154.4	10.7	11.6	4.0	395.5	126.0		4.15	6.0	134.0	EXT1300
135	4.00	+0.00 -0.10	128.0		159.8	11.0	11.8	4.0	389.5	131.0		4.15	6.0	139.2	EXT1350
140	4.00		133.0		165.2	11.2	12.0	4.0	376.5	136.0		4.15	6.0	144.5	EXT1400
145	4.00		138.0		170.6	11.5	12.2	4.0	367.0	141.0		4.15	6.0	149.6	EXT1450
150	4.00		142.0		177.3	11.8	13.0	4.0	357.5	145.0		4.15	6.0	193.0	EXT1500
155	4.00		146.0	+0.63 -1.50	182.3	12.0	13.0	4.0	352.9	150.0	+0.00 -0.63	4.15	7.5	199.6	EXT1550
160	4.00		151.0		188.0	12.2	13.3	4.0	349.2	155.0		4.15	7.5	206.1	EXT1600
165	4.00		155.5		193.4	12.5	13.5	4.0	345.3	160.0		4.15	7.5	212.5	EXT1650
170	4.00		160.5		198.4	12.9	13.5	4.0	349.2	165.0		4.15	7.5	219.1	EXT1700
175	4.00		165.5		203.4	12.9	13.5	4.0	340.1	170.0		4.15	7.5	225.5	EXT1750
180	4.00		170.5		210.0	14.0	14.2	4.0	345.3	175.0		4.15	7.5	232.2	EXT1800
185	4.00		175.5		215.0	14.0	14.2	4.0	336.7	180.0		4.15	7.5	238.6	EXT1850

Most sizes over 170mm are without lugs

Measurements in mm & kN

Shaft	Circlip Dimensions									Groove Dimensions					Code No.
	t	Tol	D	Tol	Cs	b	L	h	Fr kN	G	Tol	W	n	Fr kN	
190	4.00		180.5		220.0	14.0	14.2	4.0	333.8	185.0		4.15	7.5	245.1	EXT1900
195	4.00	+0.00 -0.10	185.5		225.0	14.0	14.2	4.0	325.4	190.0		4.15	7.5	251.8	EXT1950
200	4.00		190.5		230.0	14.0	14.2	4.0	319.2	195.0		4.15	7.5	258.3	EXT2000
205	5.00		193.0		235.0	14.0	14.2	4.0	611.7	199.0		5.15	9.0	317.5	EXT2050

210	5.00		198.0		240.0	14.0	14.2	4.0	598.2	204.0		5.15	9.0	325.1	EXT2100
215	5.00		203.0		245.0	14.0	14.2	4.0	585.1	209.0		5.15	9.0	332.8	EXT2150
220	5.00		208.0		250.0	14.0	14.2	4.0	572.4	214.0	+0.00 -0.72	5.15	9.0	340.8	EXT2200
225	5.00		213.0	+0.72 -1.7	255.0	14.0	14.2	4.0	559.8	219.0		5.15	9.0	349.1	EXT2250
230	5.00		218.0		260.0	14.0	14.2	4.0	548.9	224.0		5.15	9.0	356.6	EXT2300
235	5.00		223.0		265.0	14.0	14.2	4.0	537.1	229.0		5.15	9.0	364.1	EXT2350
240	5.00		228.0		270.0	14.0	14.2	4.0	530.3	234.0		5.15	9.0	372.6	EXT2400
245	5.00		233.0		275.0	14.0	14.2	4.0	512.2	239.0		5.15	9.0	380.0	EXT2450
250	5.00	+0.00 -0.12	238.0		280.0	14.0	14.2	4.0	504.3	244.0		5.15	9.0	388.3	EXT2500
255	5.00		240.0		289.0	16.0	16.2	5.0	557.1	247.0		5.15	12.0	525.0	EXT2550
260	5.00		245.0		294.0	16.0	16.2	5.0	540.6	252.0		5.15	12.0	535.8	EXT2600
265	5.00		250.0		299.0	16.0	16.2	5.0	536.2	257.0		5.15	12.0	546.6	EXT2650
270	5.00		255.0		304.0	16.0	16.2	5.0	525.3	262.0		5.15	12.0	556.6	EXT2700
275	5.00		260.0		309.0	16.0	16.2	5.0	516.7	267.0		5.15	12.0	566.6	EXT2750
280	5.00		265.0		314.0	16.0	16.2	5.0	508.2	272.0		5.15	12.0	576.6	EXT2800
285	5.00		270.0		319.0	16.0	16.2	5.0	499.1	277.0	+0.00 -0.81	5.15	12.0	587.5	EXT2850
290	5.00		275.0	+0.81 -2.00	324.0	16.0	16.2	5.0	490.8	282.0		5.15	12.0	599.1	EXT2900
295	5.00		280.0		329.0	16.0	16.2	5.0	481.8	287.0		5.15	12.0	609.1	EXT2950
300	5.00		285.0		334.0	16.0	16.2	5.0	475.0	292.0		5.15	12.0	619.1	EXT3000
310	6.00		293.0		352.5	20.0	20.2	6.0	1016.9.0	300.0		6.20	15.0	796.6	EXT3100
320	6.00		303.0		362.5	20.0	20.2	6.0	988.6	310.0		6.20	15.0	825.0	EXT3200
330	6.00		313.0		372.5	20.0	20.2	6.0	958.4	320.0		6.20	15.0	850.0	EXT3300
340	6.00		323.0		382.5	20.0	20.2	6.0	932.7	330.0		6.20	15.0	876.6	EXT3400
350	6.00	+0.00 -0.18	333.0		392.5	20.0	20.2	6.0	905.6	340.0		6.20	15.0	903.3	EXT3500
360	6.00		343.0		402.5	20.0	20.2	6.0	880.7	350.0	+0.00 -0.89	6.20	15.0	928.3	EXT3600
370	6.00		353.0	+0.90 -2.50	412.5	20.0	20.2	6.0	856.7	360.0		6.20	15.0	955.0	EXT3700
380	6.00		363.0		422.5	20.0	20.2	6.0	833.5	370.0		6.20	15.0	980.0	EXT3800
390	6.00		373.0		432.5	20.0	20.2	6.0	814.3	380.0		6.20	15.0	1008.0	EXT3900
400	6.00		383.0		442.5	20.0	20.2	6.0	793.4	390.0		6.20	15.0	1033.0	EXT4000